

AI Data Scientist

1. Python for Data Science and Scientific Computing

- **YouTube:**

- [Python for Data Science Full Course by freeCodeCamp](#) – Thorough dive into essential syntax, logic, and arrays.

- [Corey Schafer Python Playlist](#) – The ultimate industry baseline for object-oriented Python and clean coding practices.

- **Hands-on Labs:**

- [Kaggle Learn: Python](#) – Fast, interactive browser-based environments handling syntax structures.

- [Exercism: Python Track](#) – Code execution drills with real peer code-review loops.

- **Documentation:**

- [NumPy User Guide](#) & [Pandas Getting Started](#) – The definitive structural optimization manuals.

2. Exploratory Data Analysis (EDA) and Visualization

- **YouTube:**

- [Data Analysis with Python by Jovian](#) – Comprehensive visual guides parsing massive, dirty datasets.

- [StatQuest: Matplotlib and Seaborn Guided Maps](#) – Breaks down complex plot mechanics into simple logic.

- **Hands-on Labs:**

- [Kaggle Datasets and Notebooks](#) – Practice EDA directly on raw, open-source enterprise records.

- [DataCamp Interactive EDA Courses](#) – In-browser scripting environments for data manipulation.

- **Documentation:**

- [Seaborn Gallery Documentation](#) – High-yield visual dictionary mapping code examples to statistical graphs.

3. Feature Engineering and Data Preparation

- **YouTube:**

- [Feature Engineering for Machine Learning by CampusX](#) – Deep dive into imputation, encoding, mathematical transformations, and outlier treatment.

- [Krish Naik Feature Engineering Playlist](#) – Production-grade techniques for handling real-world data issues.

- **Hands-on Labs:**

- [Kaggle Learn: Feature Engineering](#) – Advanced pipeline scripts for constructing meaningful data traits.

- [Machine Learning Repository Data Challenges \(UCI\)](#) – Download raw data matrices to manually run scaling and normalization routines.

- **Documentation:**

- [Scikit-Learn Preprocessing Guide](#) – Mathematical explanations behind data scaling, encoders, and vectorization methods.

4. Model Evaluation and Experimentation

- **YouTube:**

- StatQuest: ROC, AUC, Bias and Variance – Intuitive visual breakdowns of classification metrics and validation rules.

- Valerio Velardo - The Sound of AI (Model Evaluation) – Comprehensive view of cross-validation architectures and avoiding data leakage.

- **Hands-on Labs:**

- [Neptune.ai Sandbox Overviews](#) – Connect model metrics directly to operational experiment tracking boards.

- [Weights & Biases \(W&B\) Walkthroughs](#) – Live dashboard exercises mapping validation loops and loss curves.

- **Documentation:**

- [Scikit-Learn Model Evaluation Matrix](#) – Comprehensive registry covering precision, recall, F1, and custom metrics.

5. Supervised Learning Algorithms and Modeling

- **YouTube:**

- [Machine Learning Zoomcamp by Alexey Grigorev](#) – The definitive software engineering approach to building regression and tree algorithms.

- [StatQuest Machine Learning Video Series](#) – Step-by-step visual proofs of SVMs, Random Forests, and Gradient Boosting.

- **Hands-on Labs:**

- [Scikit-Learn Algorithm Sandboxes](#) – Native python execution scripts simulating classification algorithms.

- [Kaggle Competitions \(Titanic / House Prices\)](#) – The baseline playground for engineering modeling solutions.

- **Documentation:**

- **Scikit-Learn User Manual** – The industry standard blueprint for classical machine learning architectures.

6. Unsupervised Learning and Representation Learning

- **YouTube:**

- **Stanford CS229: Machine Learning by Andrew Ng** – Academic, rigorous mathematical breakdowns of clustering and PCA logic.
- **Luis Serrano Clustering Frameworks** – Geometric, intuitive guide explaining K-Means and Hierarchical Clustering.

- **Hands-on Labs:**

- **UCI K-Means & PCA Practical Notebooks** – Scripts modeling segmentation, customer profiling, and matrix reduction.
- **Google Developers ML Clustering Course** – Guided practice modules modeling data similarity matching.

- **Documentation:**

- **Scikit-Learn Unsupervised Modules** – Comprehensive technical specifications for manifold learning and decomposition.

7. Deep Learning for Structured and Unstructured Data

- **YouTube:**

- **Deep Learning Specialization by Andrew Ng (DeepLearning.AI)** – The structural baseline covering Neural Networks, CNNs, and RNNs.
- **PyTorch for Deep Learning Course by Daniel Bourke** – Purely practical 25-hour course covering computer vision and tensor manipulation.

- **Hands-on Labs:**

- **TensorFlow Playgrounds** – Visual browser simulation showing neural networks adjusting weights in real-time.
- **Fast.ai Practical Deep Learning for Coders** – Top-down engineering approach to deploying production-grade deep networks.

- **Documentation:**

- **PyTorch Documentation Tutorials** – Clean, pristine blueprints for custom network architectures and data loaders.

8. MLOps, Deployment, and Production Engineering

- **YouTube:**

- **MLOps Zoomcamp by DataTalksClub** – Absolute gold standard implementation series tracking ML pipelines from local code to cloud production.
- **TechWorld with Nana MLOps Overview** – Architectural mapping of model registries, tracking servers, and pipeline tools.

- **Hands-on Labs:**

- GCP / AWS MLOps Workshops – Sandbox configurations for building automated CI/CD pipelines via SageMaker or VertexAI.
- MLflow Guided Integration Scenarios – Spin up tracking servers to store hyperparameters and model artifacts locally.

- **Documentation:**

- The MLOps Roadmap (Community Standards) – Production reference manual defining artifact lineage, feature stores, and automated data monitoring.

9. Experiment Design, Decision Making, and AI Systems

- **YouTube:**

- A/B Testing Course by Udacity (Google Engineers) – Essential training covering hypothesis parameters, sample sizing, and statistical power.
- MIT 6.S191: Introduction to Deep Learning (AI Systems) – Architecture design reviews centered on building scalable, safe, and fair AI platforms.

- **Hands-on Labs:**

- Evan Miller's A/B Testing Toolsets – Statistical workspace simulators designed for plotting sample sizes and validation boundaries.
- Google What-If Tool Interactive Demos – Sandbox probing fairness constraints and adversarial changes in data behavior.

- **Documentation:**

- Microsoft Responsible AI Principles Guide & Google AI Principles – Strict enterprise design specifications for building unbiased model architectures.